

Learnable optical model for hologram correction

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Objective

- The quality of holograms often suffer from experimental and simulation mismatch
- This mismatch is due to imperfections along the optical path
 - Wavefront distortion, misalignment, nonideal SLM response, etc.

How can we use computational methods to correct for the observed experimental and simulation mismatch?

Introduction to Holography

- *Holography*
 - Technique where wavefronts can be recorded then reconstructed
- *Computer-generated holography (CGH)*
 - Apply computer algorithms to generate or prepare holograms
- *Metasurface holography*
 - Use of subwavelength nanostructures
 - Design geometry of nanoposts to shape light wavefront for hologram generation [1],[2],[3]
- Applications: AR/VR display, medical imaging, optical tweezers, **3D printing**, etc. [4],[5]

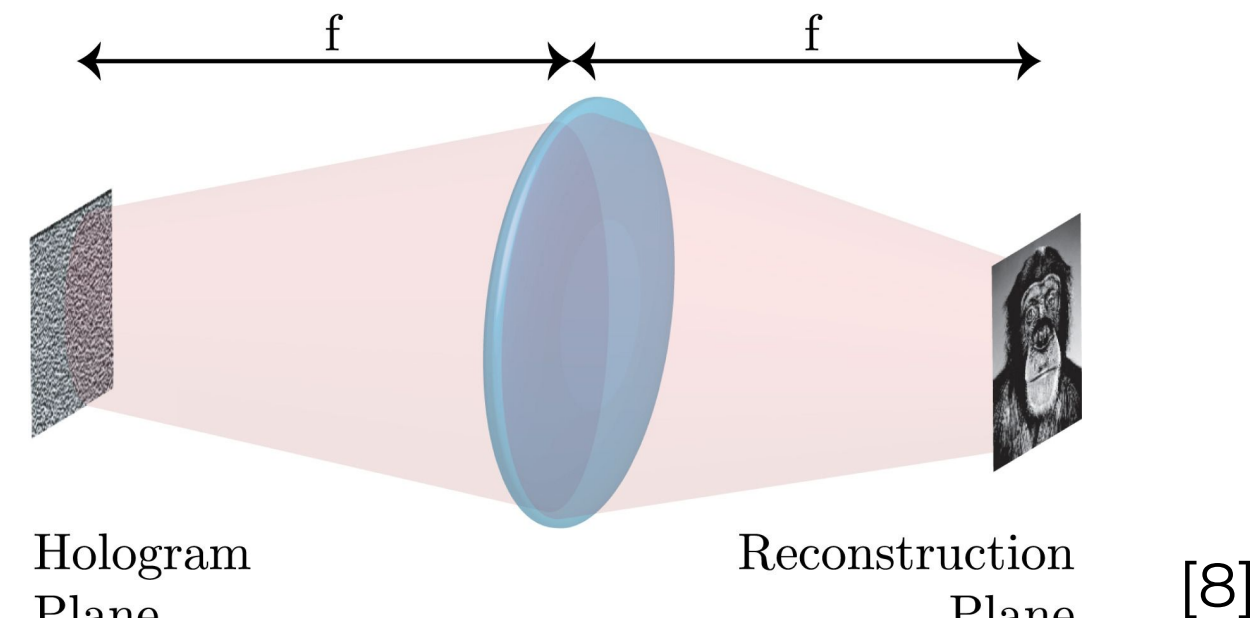


Figure 1: Holography optical setup [8]

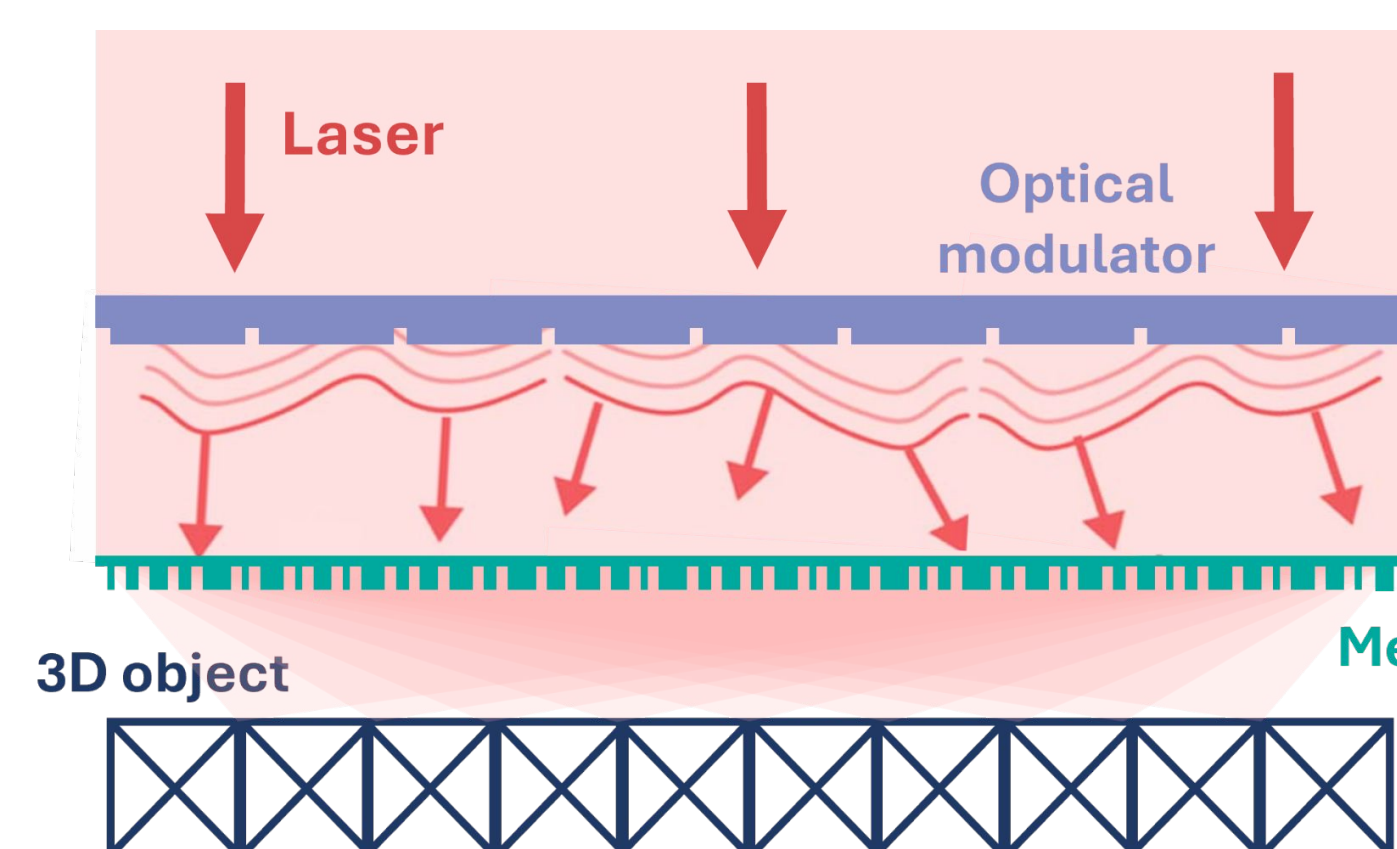


Figure 2: Printing setup with SLM & Metasurface

Training & Alignment Methodology

- Train model
 - 2k dataset
 - Shapes
 - Borders
 - Images

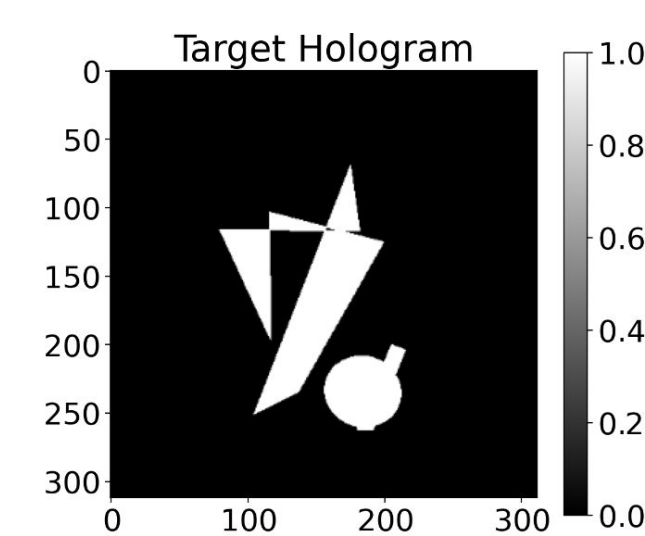


Figure 3: Dataset example (random shapes)

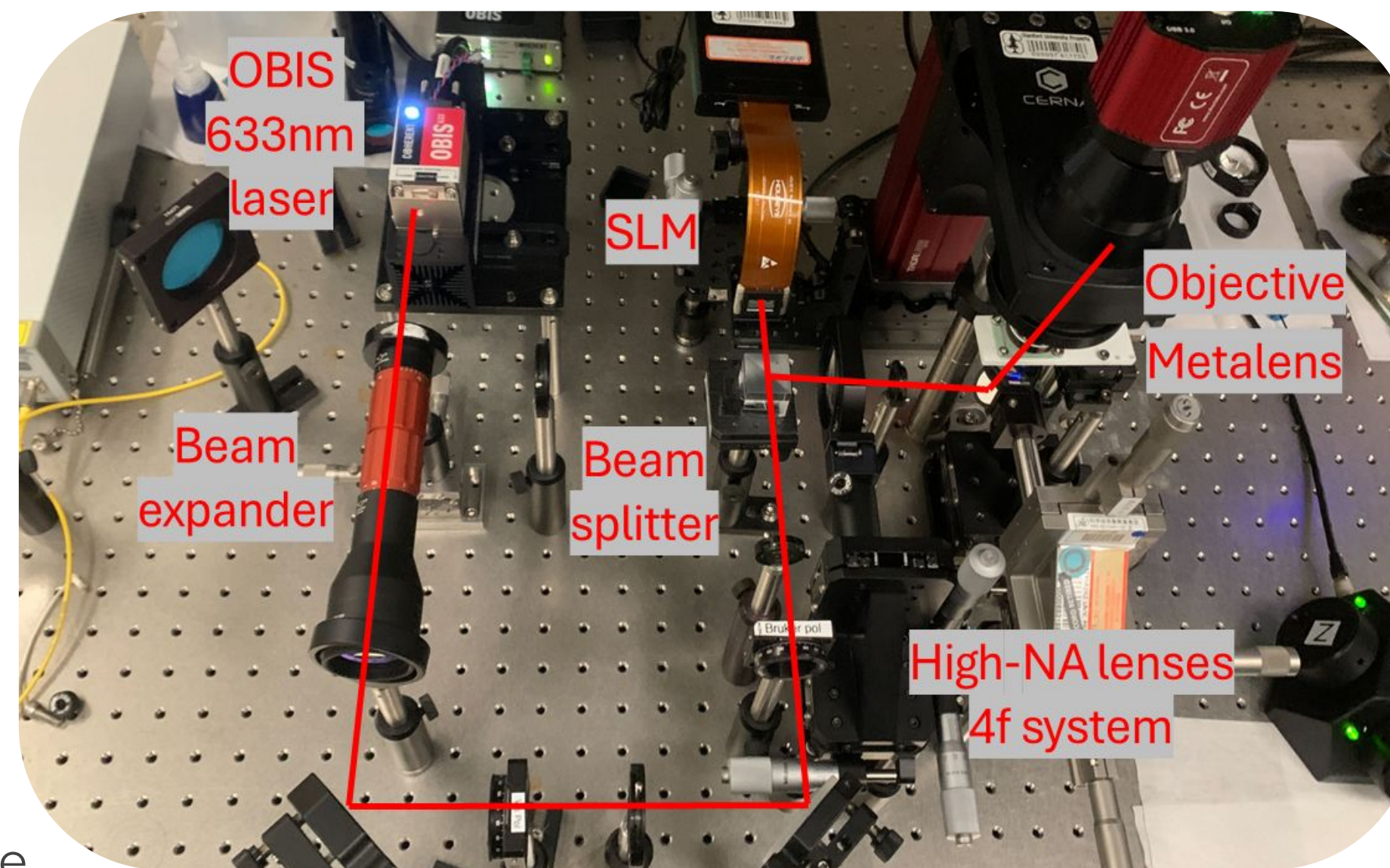


Figure 4: Experimental setup

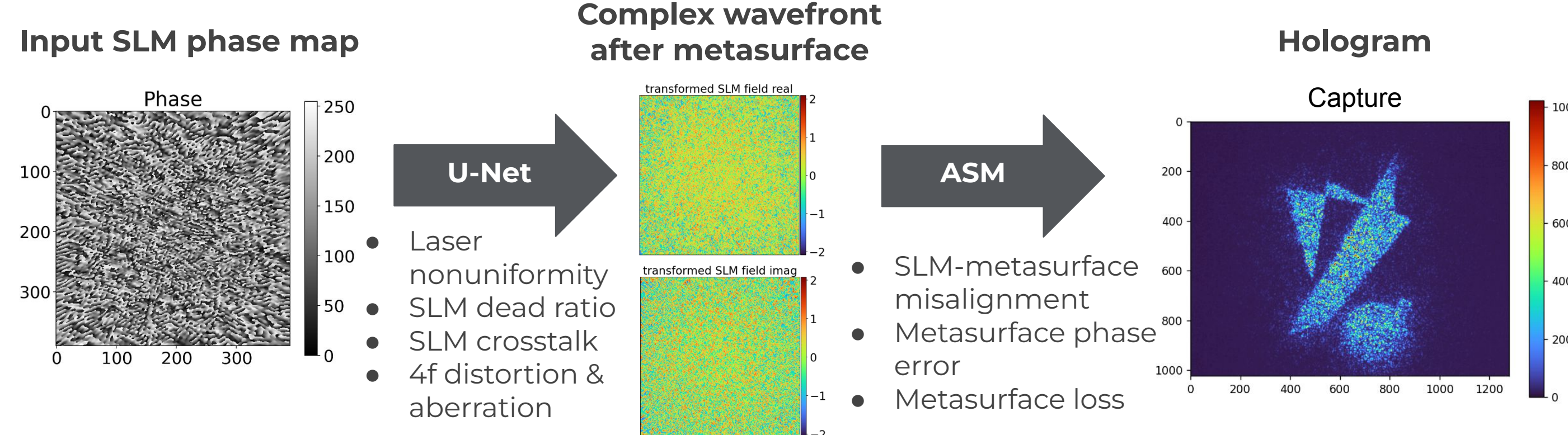


Figure 5: Algorithm progression and potential errors

- Camera-in-the-loop (CITL)

- Captured hologram is compared with simulated hologram [6],[7]
- Update phase in direction to minimize loss

$$\min_{\phi} \mathcal{L}(|\hat{f}(\phi)|, a_{\text{target}})$$

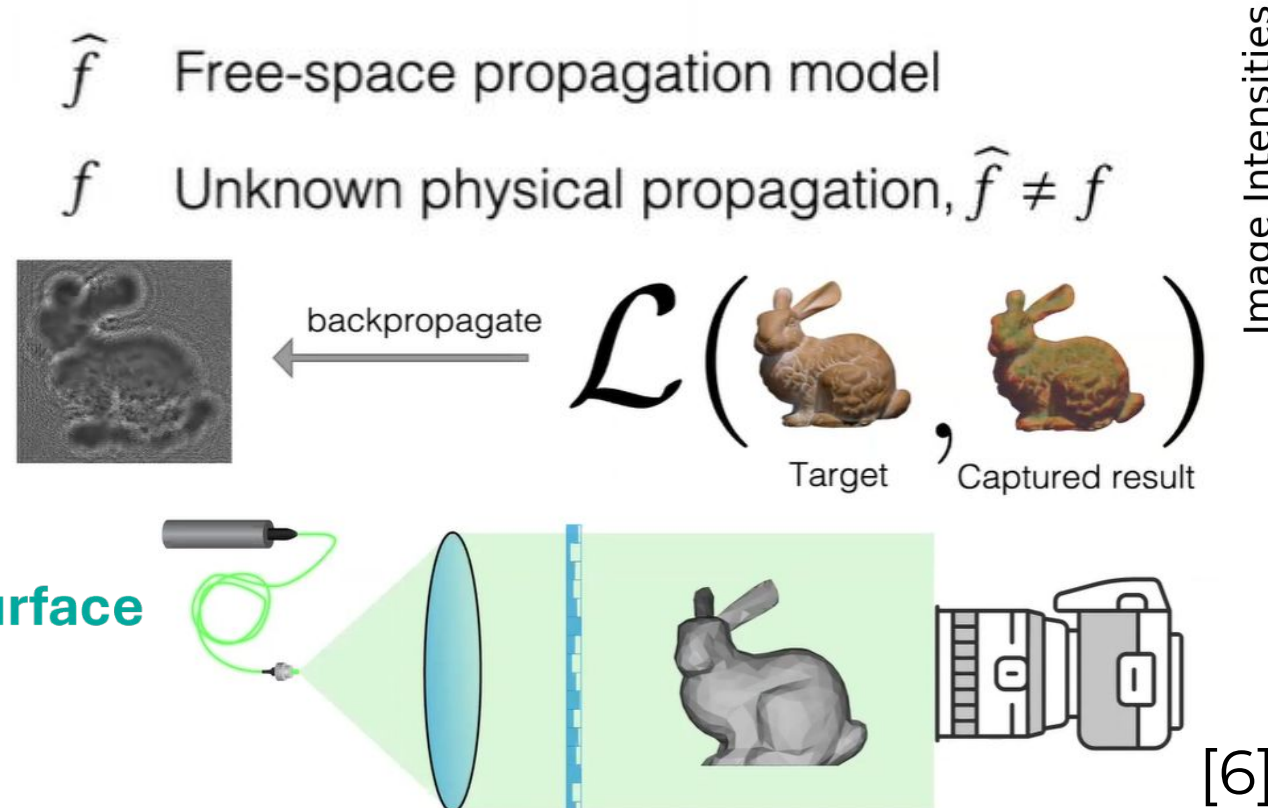


Figure 6: CITL approach

- Autofocus

- Align XY planes with random dots
- Find ideal Z plane at peak image intensity

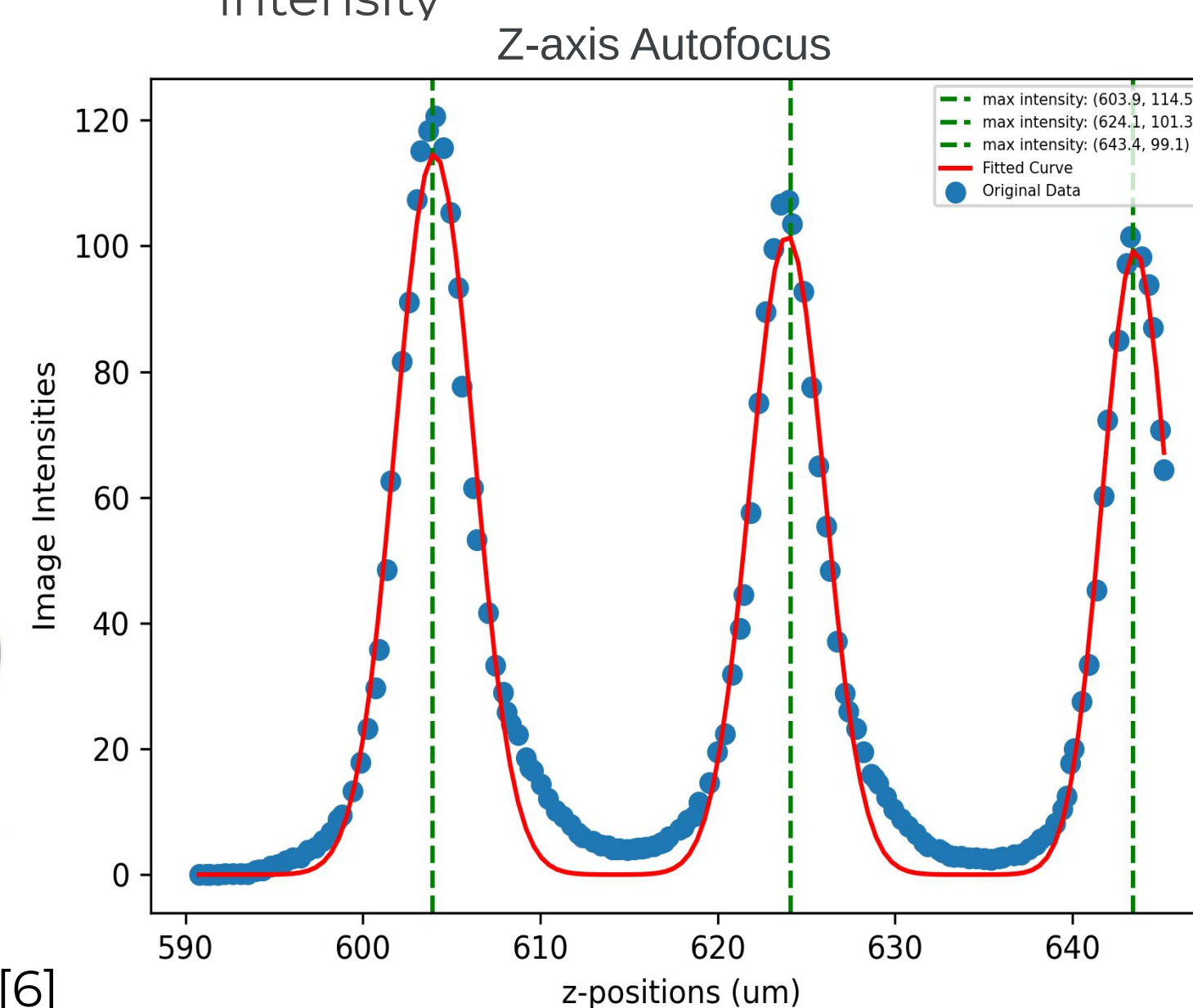


Figure 7: 3D hologram autofocus plot

Learnable Optical Model Results

- Trained model with CITL and XYZ autofocus performs the best
- Trained model & CITL account for physical characteristics
 - Correct for residual error not captured in pure simulation
- CITL model can learn speckle (grainy, noise-like patterns) and correct for it
- Partially corrects for zeroth-ordered light (non-diffracted light)

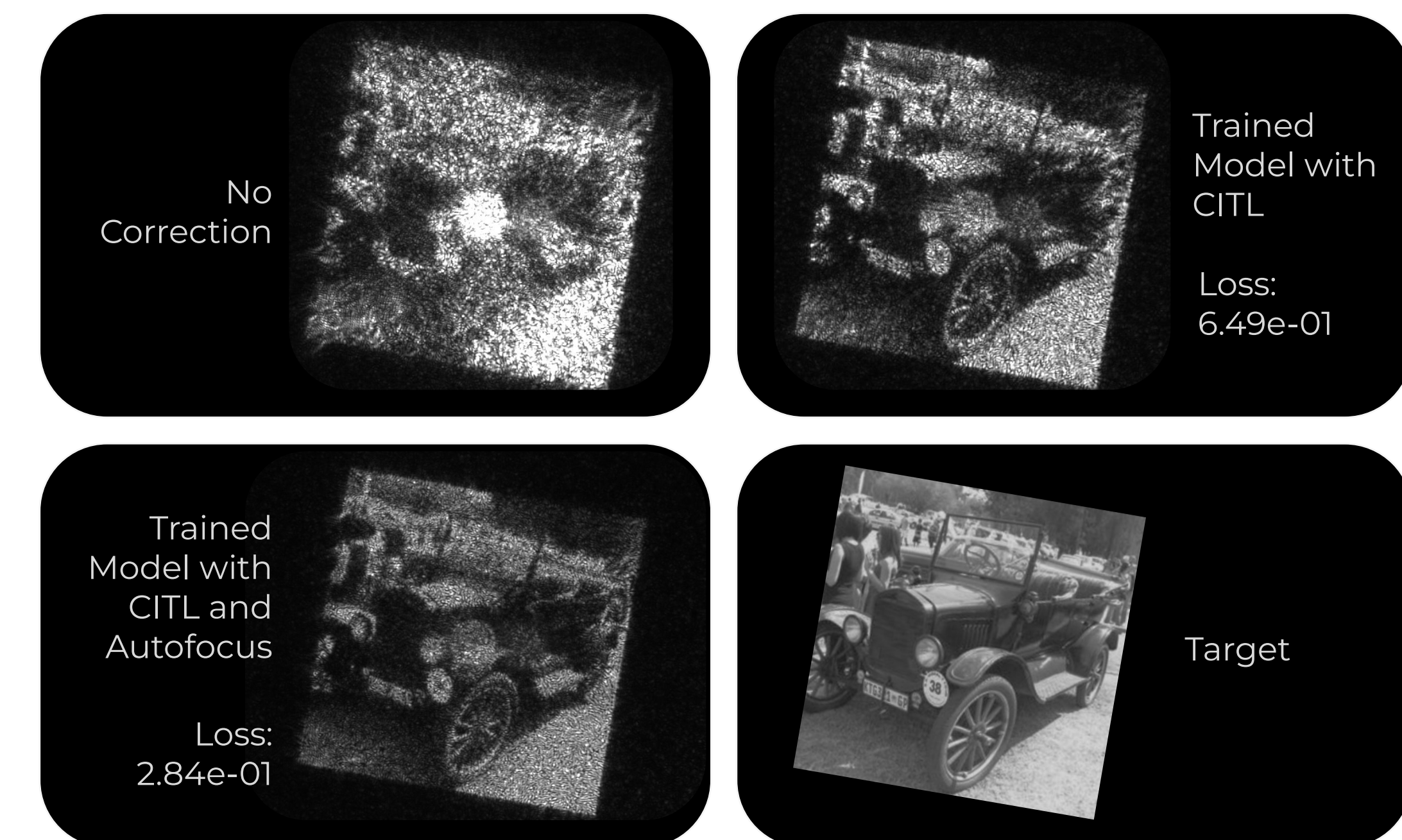


Figure 8: Comparison of hologram captures

Future Directions

- Extend model to account for different types of metasurfaces
- Add learning rate optimization for learned model and CITL
- Train model for 3D holograms

Acknowledgements & References

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